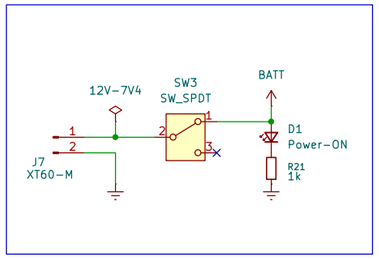
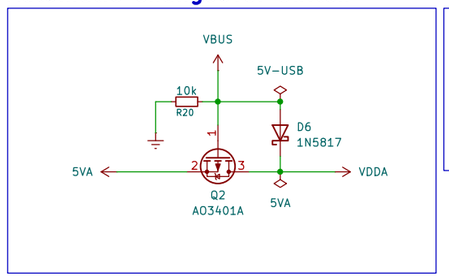
Power Supply

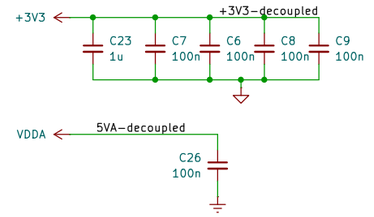


Automatic changeover switch



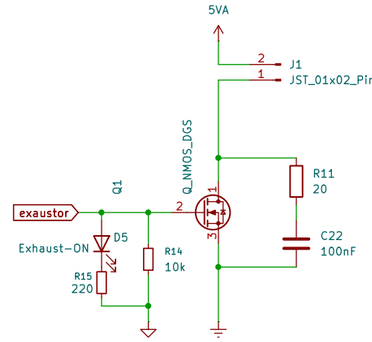
- When the external 5V (Vbus) is connected, the internal 5V (buck) is disconnected to MCU.
- When the external 5V (Vbus) is disconnected, power is supplied by the internal 5V (buck).

MCU Decoupling



100nF capacitor decoupling per pins and 1uF for high density power supply

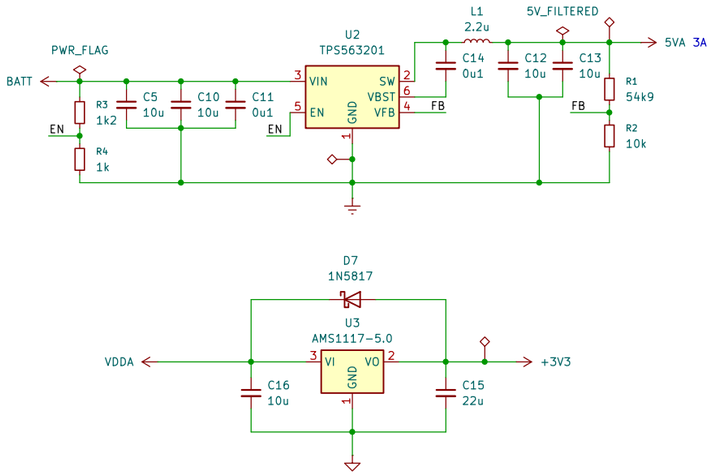
Exhaust



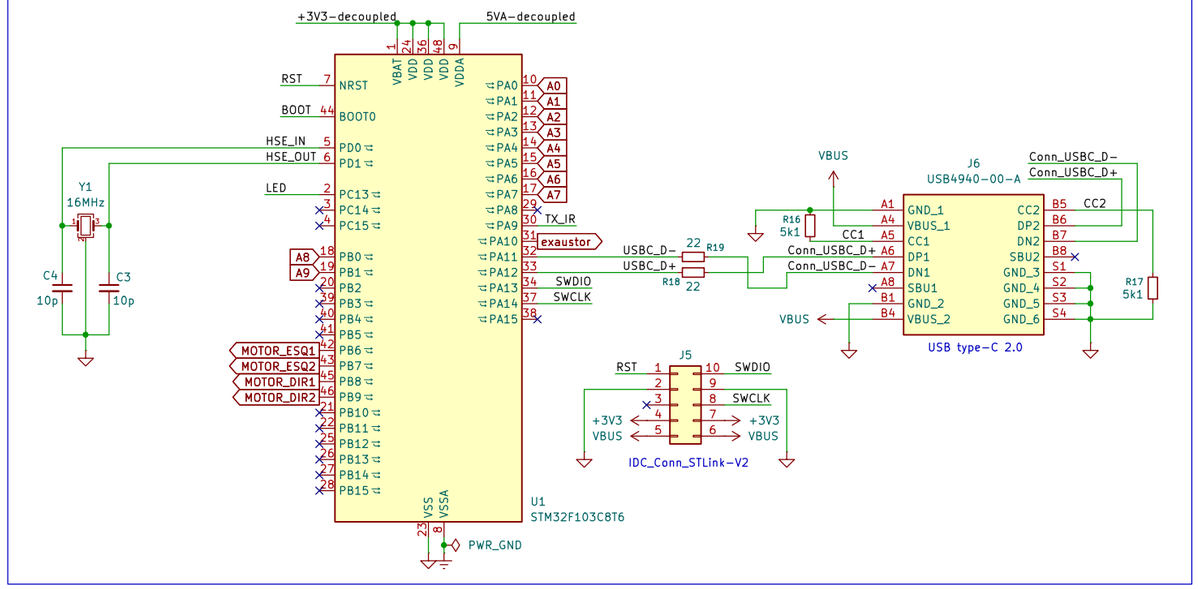
Electronic switching of an exhaust fan motor using a MOSFET, with an implemented snubber filter.

Power Rails (5V/3V3)

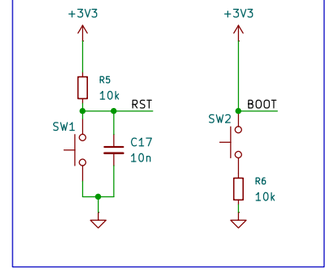
Buck converter and LDO implementation



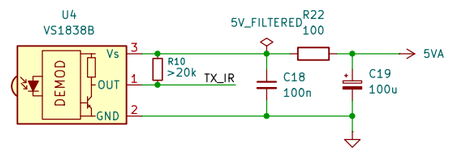
MCU



Hardware Reset and Boot Mode



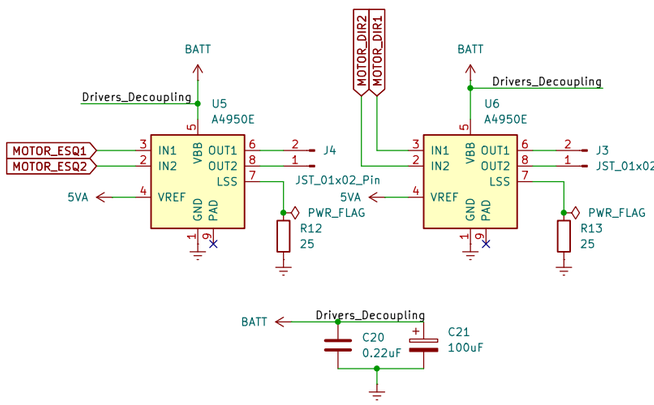
IR Receiver



Sensor sensitive to voltage variations (100µF capacitor) and current-limiting resistor (100 Ω)

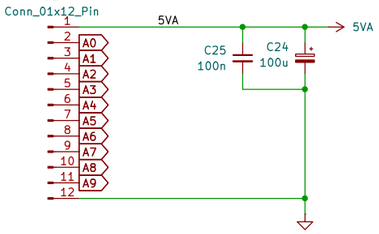
Note: The pull-up resistor must be set to a value greater than 20kΩ.

Motor Drivers

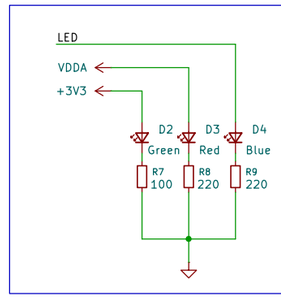


25Ω resistor for maximum current configuration (datasheet)

Array-sensors connector



Debug LEDs



Features

- Exhaust (LAIO)
- Sensor DIY (KAUÊ)
- Wheel DIY
- A4950 (LAIO)
- Motor FK-132H
- VS1838B (KAUÊ)

Sugestions

- Buzzer NO
- LED to Exhaust YES

Kauê Lucas Nascimento da Silva & Laio

Sheet: /
File: HSLF.kicad_sch

Title: HSFL MotherBoard

Size: A3	Date: 2026-07-16	**Rev: 1.1**
KiCad E.D.A. 10.0.1		Id: 1/1